Association Between Mental Health and Overweight/Obesity Among Reproductive Women in Chattogram Division: A Cross-Sectional Study

NUTRITION SUMMIT

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Background

- ❖ Emerging evidence links with Depression, anxiety and stress (DAS) to higher rates of overweight and obesity[1]
- ❖ Depression, anxiety, stress (DAS) disorders are rising globally, affecting physical health[2]
- ❖ The prevalence of overweight and obesity of women (28%) are more likely than males (22%), in 2013[3]
- ❖ Peripartum depression ranges from 6.5%-13% in high-income countries and 16%-20% in low- and middle-income countries[4]
- ❖ Overweight/obesity is the risk factor of chronic diseases[5]

Objective



This study aims to investigate the association between mental health disorders - depression, anxiety, and stress and overweight or obesity among reproductive aged women in the Chattogram division.

Methodology

Study design	Cross-sectional
Questionnaire	DASS-21
Data Source	Reproductive Women
Sample size	405
Study Area	Chattogram division
Data Analysis	Chi-square, Binary logistic regression, Hosmer-Lemeshow goodness-of-fit test

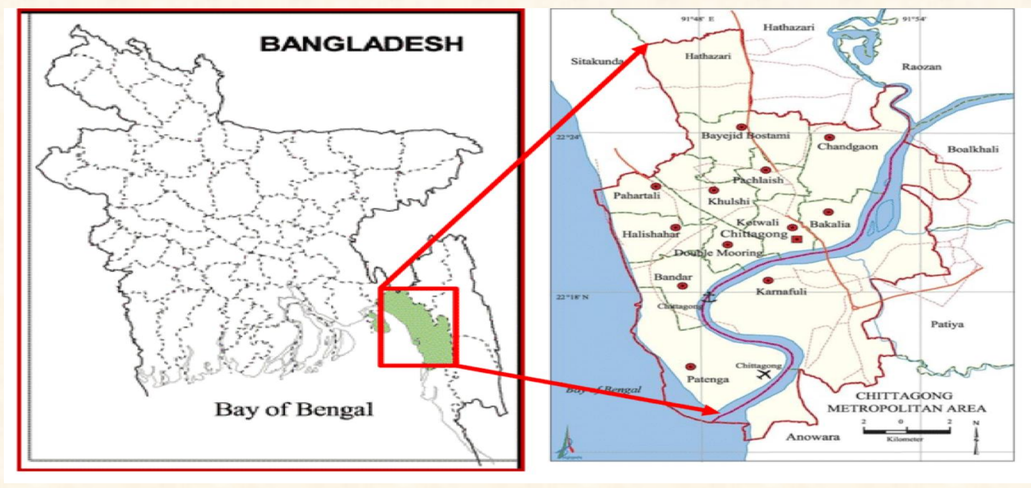


Figure 1: Map of Study Area (Chittagong metropolitan area)

Result & Findings

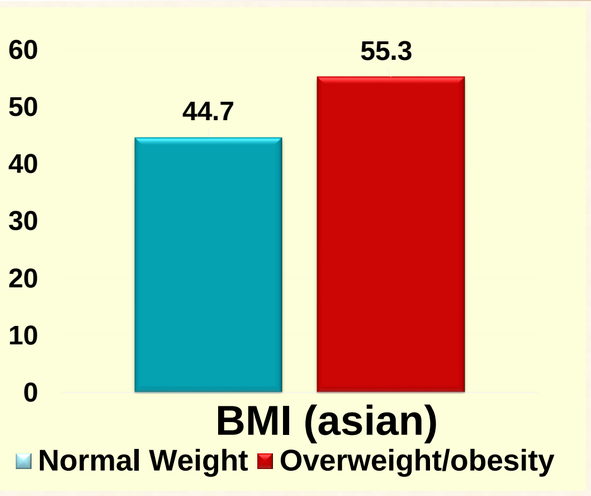


Figure 2: BMI Status

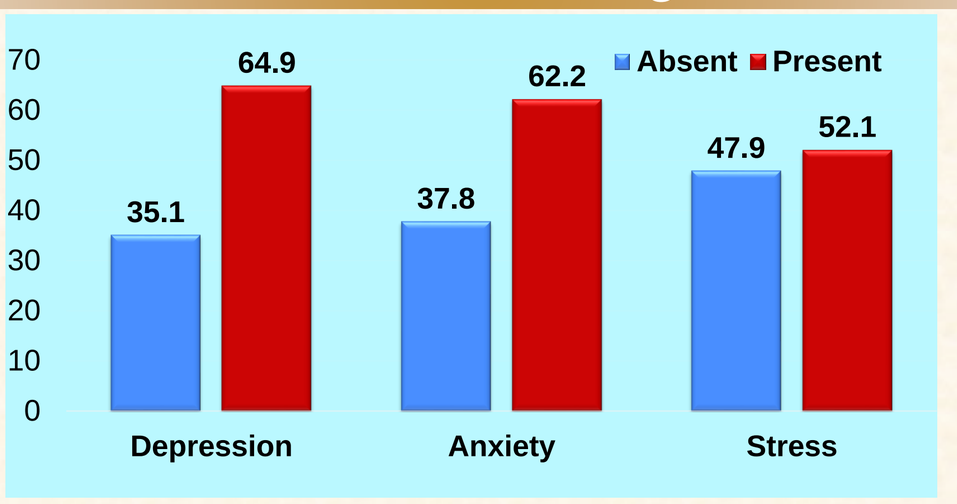


Figure 3: Status of Depression, Anxiety and Stress

Variable		Adjusted Model		
		OR	95%CI	P value
Age	15< 30	Rf		
	30<45	1.497	0.829 - 2.701	0.181
	45 or above	3.887	1.393 – 10.846	0.01
Occupation	Student	Rf		
	Unemployed	4.567	1.968 – 10.601	0.000
	Employed	1.479	0.733 – 2.984	0.274
Depression	Absent	Rf		
	present	1.776	1.142 – 2.762	0.011

Table 1: Association between depression,sociodemographic variables and overweight/obesity

- ❖ 44.7% of participants had a normal BMI, while 55.3% were overweight or obese, indicating a higher prevalence.
- ❖ The study found high mental health disorder rates: 64.9% had depression, 62.2% anxiety, and 52.1% stress, highlighting widespread challenges.
- ❖ After adjusting of sociodemographic variables women with depression were 77.6% more likely to be overweight or obese (OR 1.776, 95% CI: 1.142–2.762), showing a strong association.
- ❖ These findings suggest depression significantly contributes to overweight and obesity, while anxiety and stress have no direct impact.

Conclusions

This study highlights the association between depression and obesity in reproductive-aged women, urging integrated interventions for both mental and physical health.

Future research should explore the mechanisms behind this link and develop strategies to prevent and manage depression, reducing obesity and related health risks.

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Background: The prevalence of mental health disorders, including depression, anxiety, and stress (DAS), are increasing globally. As mental and physical health are interconnected, DAS disorders are hypothesized to be associated with overweight and obesity.
Objective: This study aims to determine the association between mental health disorders and overweight or obesity among women in the Chattogram division.
Methods: A cross-sectional survey was conducted with 405 reproductive-aged women from the Chattogram divisions. Mental health disorders were assessed using the 21-item Depression, Anxiety, and Stress Scale (DASS-21), while overweight and obesity were classified using Asian BMI cut-off points. Descriptive and inferential statistics were employed to analyze associations between overweight/obesity, mental health disorders, and sociodemographic variables.
Results: Among the participants, 44.7% had a normal (BMI 18.5–22.9 kg/m²), while 55.3% were classified as overweight or obese (BMI ≥23 kg/m²). Depression affected 64.9% of the participants, anxiety 62.2%, and stress 52.1%. Age, marital status, occupation and depression were significantly associated with overweight/obesity. After adjusting for age, marital status, and occupation, women with depression were found to have a 77.6% higher risk of being overweight or obese compared to those without depression (OR 1.776; 95% CI, 1.142–2.762). No significant associations were observed between anxiety or stress and overweight/obesity.
Conclusion: Overweight and obesity are major contributors to non-communicable diseases. As depression was significantly associated with overweight and obesity, targeted interventions are needed to address and prevent depression to mitigate its impact on physical health.
Keywords: Overweight, Obesity, Depression, Anxiety, Stress.

Iodine Nutrition Status and Thyroid Autoimmunity During Pregnancy

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Background: Thyroid autoimmunity is a common concern among pregnant women, characterized by thyroid auto-antibodies that can lead to complications such as preterm birth, miscarriage, and neurodevelopmental defects. Iodine, an essential micronutrient for thyroid function, influences thyroid health, but the exact relationship between iodine intake and thyroid autoimmunity remains unclear. This study investigates these associations in Shanghai, where pregnant women historically face mild iodine deficiency risk.
Objective: To examine the relationship between iodine nutrition status and thyroid autoimmunity in a large cohort of pregnant women, with a focus on identifying iodine intake's impact on thyroid health.
Methodology: This cross-sectional study included 4,635 pregnant women from Shanghai. Data